

Production Planning, Plan-vs-Actual & Capacity Optimization

Industrial Analytics Portfolio Project

Project objective: Build a management-ready analytics solution for production planning, capacity utilization, machine performance, downtime and production variance.

Business context: The project models a manufacturing environment where production targets depend on machine availability, manpower, material readiness and downtime.

Key KPIs

KPI	Portfolio Dataset Result
Total Planned Production	4229778
Total Actual Production	3785322
Plan Achievement %	89.49
Production Loss	444456
Avg Machine Utilization %	90.12
Avg Capacity Utilization %	85.72
Total Downtime Hours	3954.52
Below Target Records %	24.38

Methodology

Data preparation → validation → Plan-vs-Actual analysis → machine/capacity analysis → downtime/loss analysis → management action framework.

Dashboard Design

Executive Overview; Plan vs Actual; Machine & Capacity Analysis; Downtime & Production Loss; Management Action View.

Portfolio Interpretation

The project is intended to demonstrate how operational data can be transformed into actionable management information. Areas with persistent below-target achievement, high downtime or low utilization should be prioritized for root-cause investigation and corrective action.

Resume Positioning

Developed an end-to-end manufacturing analytics solution to monitor Plan-vs-Actual production, machine utilization, capacity utilization, downtime and production losses; built KPI dashboards and machine-level analysis to identify bottlenecks and support data-driven production planning.

GitHub Note

The repository contains a simulated dataset, Excel model, Power BI dashboard specification, Python analysis template and this project report. No confidential employer information is used.